Evolution of Cosmic Structure

Lecture 14 - Reionization

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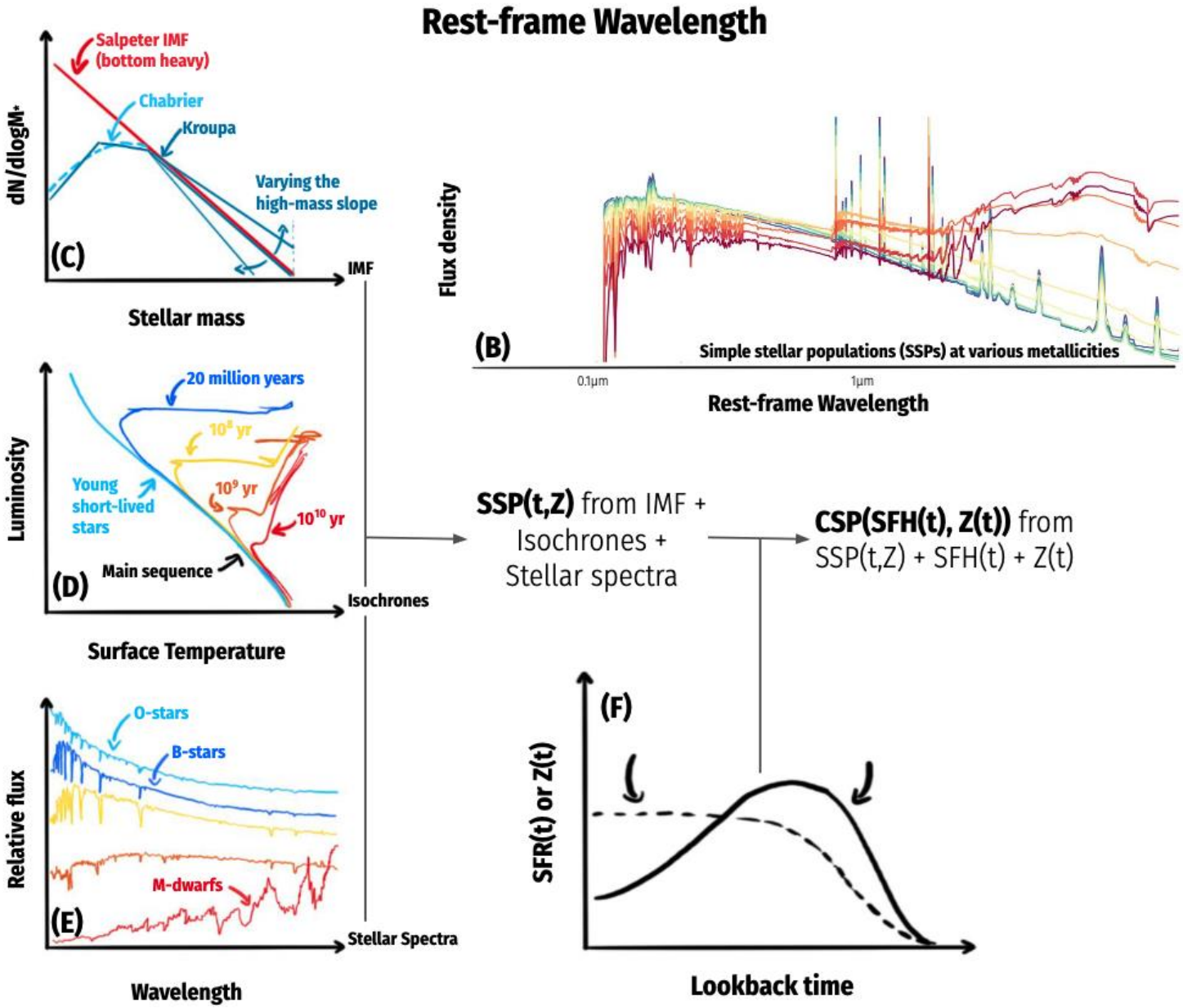
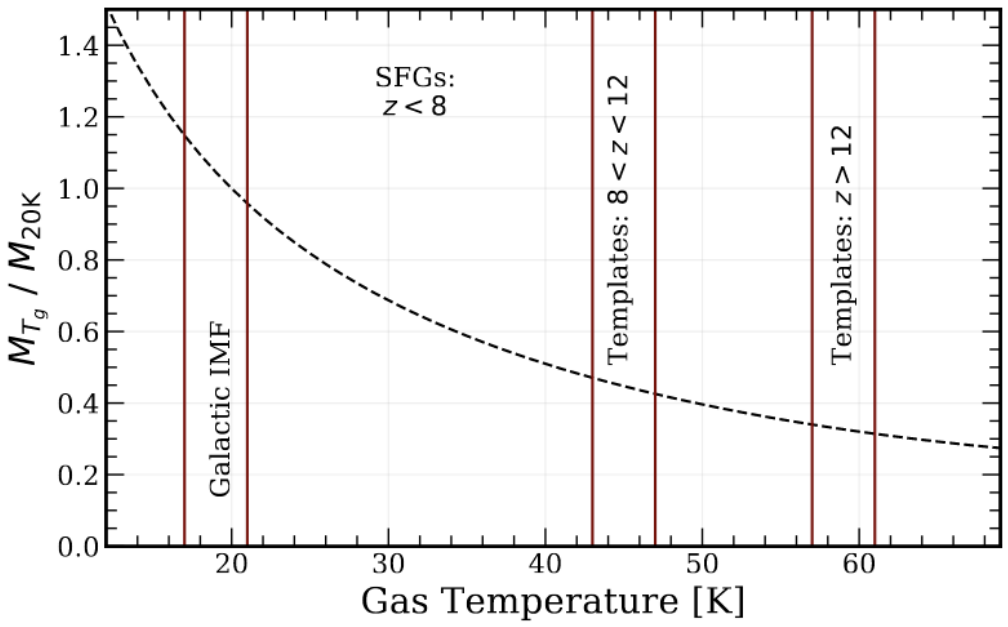
The spectral energy distribution of galaxies

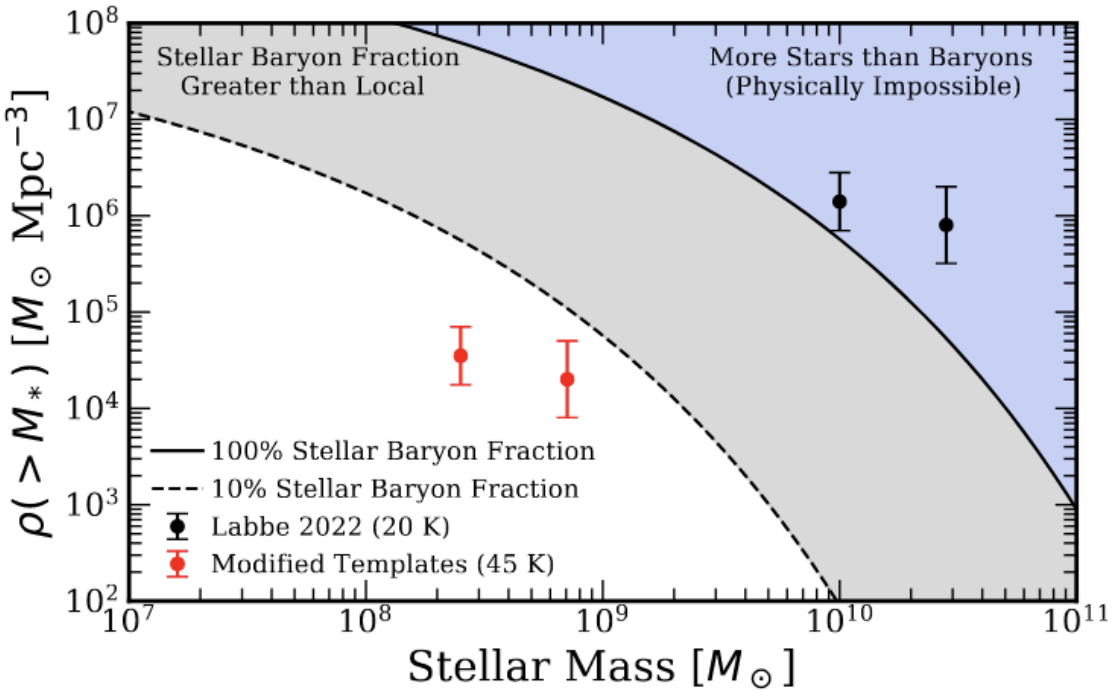
Need to exploit the SED of galaxies to find high redshift ones.



(Iyer et al. 2025)

Requirement for ‘new’ physics?





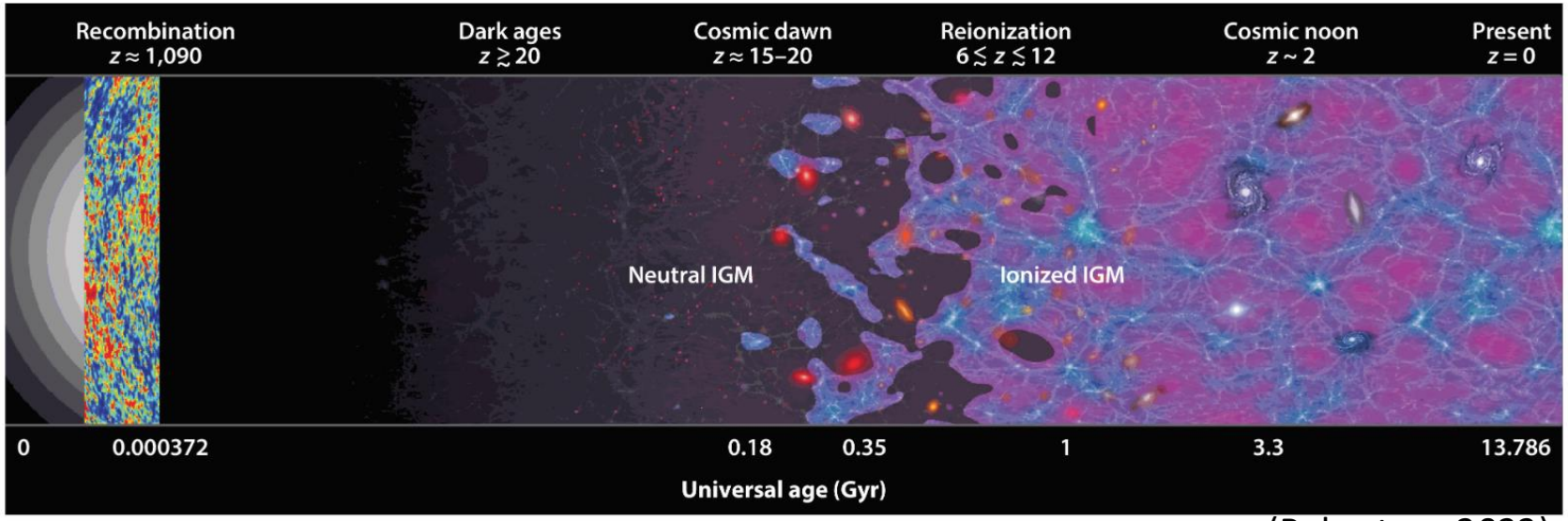
Steinhardt et al. 2022

Learning Outcomes

What do we know of the first galaxies and the history of cosmic star formation? What effect has the development of galaxies had on the rest of the baryonic matter?

- Observations of high redshift galaxies
- Reionization – the impact of star formation on the intergalactic medium
- The history of star formation and the growth of stellar mass in the Universe

Stages of ionization in the Universe



(Robertson 2022)

Recombination (z = 1100; 380,000 years after beginning)

- Electrons and protons combine – Neutral atoms – CMB 'released'

'Dark Ages'

- No sources of light, neutral atoms dominate

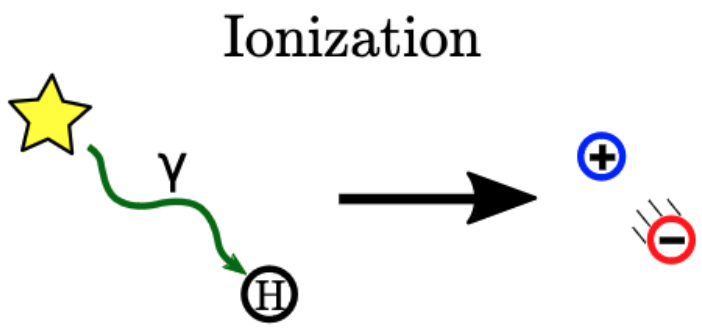
Epoch of Reionization (z = 20 – 6)

- First stars, galaxies and quasars emit ionizing radiation
- Large-scale ionized bubbles form, percolate, and fully ionize the IGM
- Universe stays mostly ionized until today (outside galaxies)

We will look at:

Energetics of reionization
Sources of reionization
Timing of reionization
Effect on small galaxies

Energetics of reionization



Recall the hydrogen atom – proton and electron interactive via the Coulomb Potential. The Schrodinger equation:

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\,\pi\varepsilon_0 r}\right]\psi = E\,\psi$$

Which gives the energy eigenvalues of

$$E_n = -\frac{m_e e^4}{8\varepsilon_0^2 h^2}\frac{1}{n^2}$$

Where n are the principal quantum numbers.

So, n = 1 is 13.6 eV. The energy needed to remove the electron from the ground state – this is the ionization energy needed.

Similarly, for neutral helium (He I → He II), the ionization energy is 24.6 eV
And for singly ionized helium (He II -> He III), the ionization energy is 54.4 eV

Helium reionization exists too, but it's hydrogen reionization which is universally meant when talking about 'reionization'

Energetics of reionization

What is the energy density required for reionization?

Just the number density of hydrogen atoms times the ionization energy (ignoring helium).

The number density is: $n_H = f_H \frac{\rho_b}{m_p}$

Where f_H is the hydrogen mass fraction (~ 0.76), ρ_b is the baryon mass density and m_p is the proton mass.

The baryon mass density is: $\rho_b = \Omega_b \rho_c = \Omega_b \frac{3 H^2}{8 \pi G}$

and the Planck 2018 results give $\Omega_b = 0.048$. Calculating for $H_0 = 67$ km/s/Mpc gives: $n_H = 1.9 \times 10^{-7} (1+z)^3 cm^{-3}$

So, the energy density is

$$U_{ion,H} = 13.6\ eV \times n_H = 4 \times 10^{-6} (1+z)^3\ eV\ cm^{-3}$$

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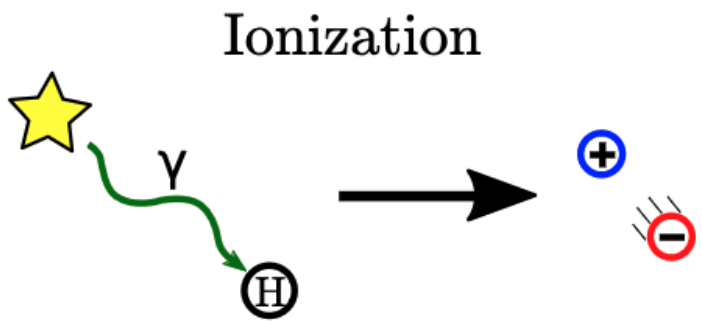
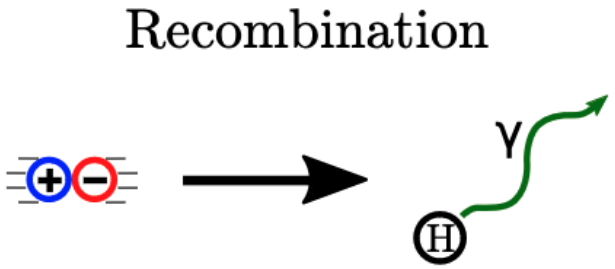
Energetics of reionization





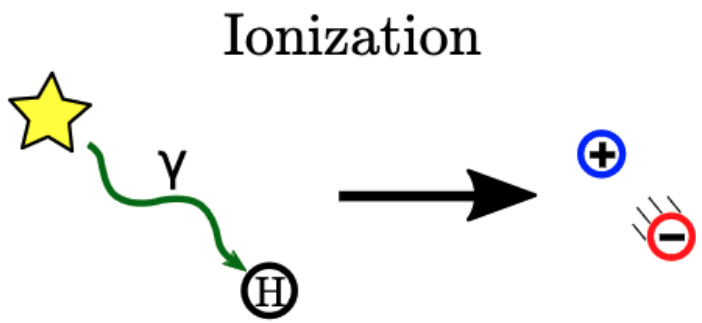
What about the recombination of electrons and protons?

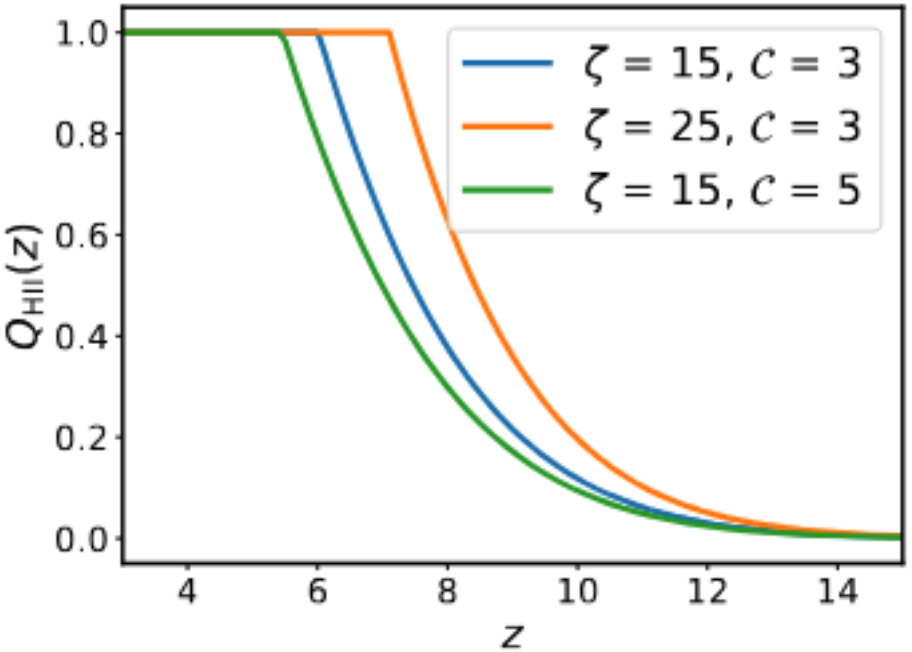
The relevant recombination rate here is known as 'Case B recombination'.

Case A: All recombination (including those directly to the n=1 ground state) contribute to the relevant recombination rate.

Case B: Only recombinations that are not directly to n=1 contribute. E.g., only recombinations to n > 1 are considered. (And lead to observable lines). This is relevant for 'optically thick nebulae' like those relevant for reionization, because the n=1 photon are immediately absorbed and reionize a nearby atom.

Energetics of reionization





(Chakraborty et al. 2025)

How can we track the recombination and ionization balance?

$$\frac{dx_e}{dt} = \frac{\dot{n}_\gamma}{n_H} - \alpha_B C_{HII} n_e x_e$$

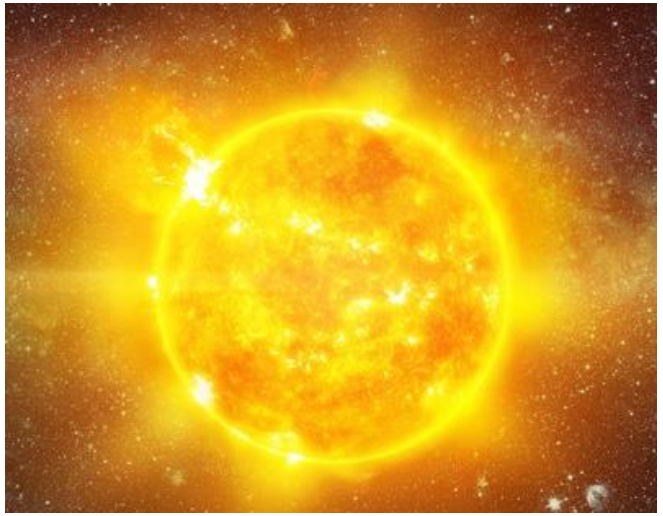
x_e is the ionized fraction, $\dot{n}_\gamma$ is the ionizing photon production rate per volume, α_B is the case B recombination coefficient, C_{HII} is the clumping factor.

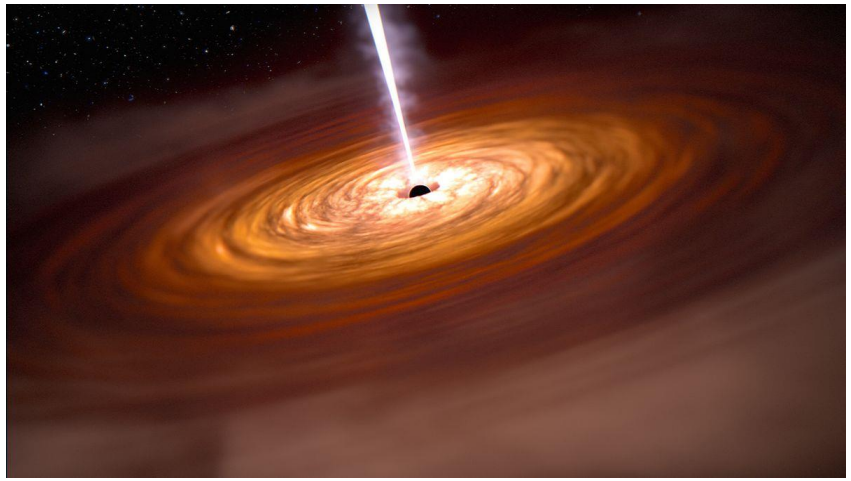
The clumping factor is defined as $C_{HII} = \frac{\langle n_e^2 \rangle}{\langle n_e \rangle^2}$

If the IGM is homogeneous then $n_e = \langle n_e \rangle$ and $C_{HII} = 1$.

A 'clumpy' IGM give $C_{HII} > 1$, so higher clumping leads to increased recombination, increasing the need for ionizing photons.

Sources of reionization





We want to find the source of the ionizing photons $\dot{n}_\gamma$

They must have energies greater than 13.6 eV → 91.2 nm $\left(E = \frac{hc}{\lambda}\right)$

First generation of stars (Population III)

- Metal-free, massive (10 – 1000 Msolar), short-lived.
- Very efficient at producing UV photons

Quasars & Active Galactic Nuclei (AGN)

- Supermassive black holes at high redshift
- Are there enough?
- Likely important for HeII reionization

Early Galaxies

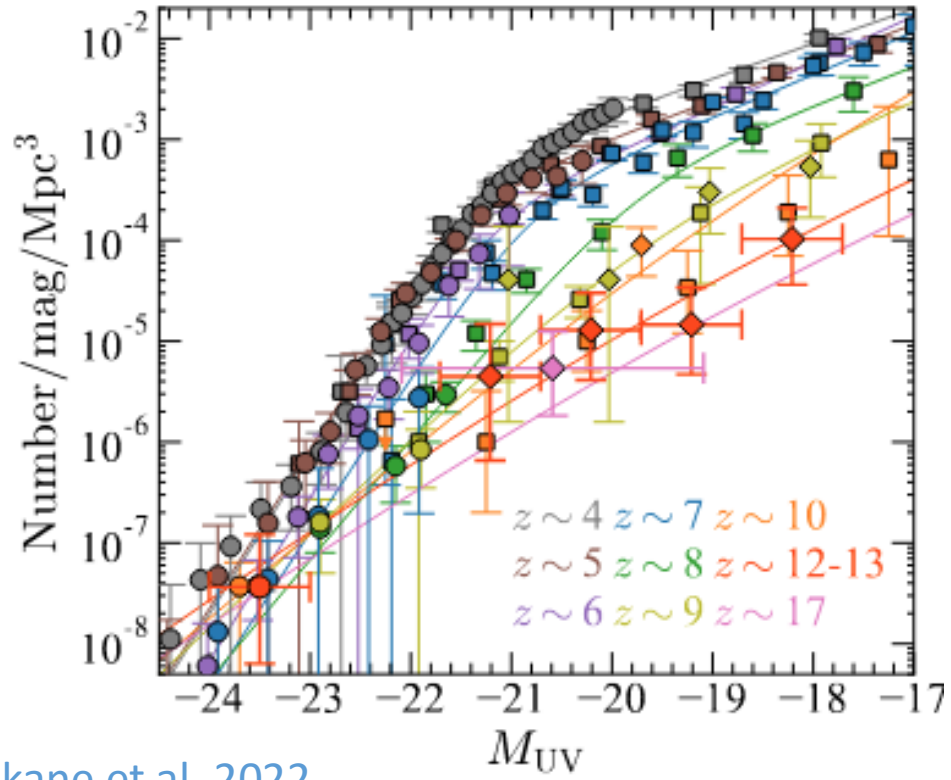
- High star formation rates
- Abundant

Sources of reionization

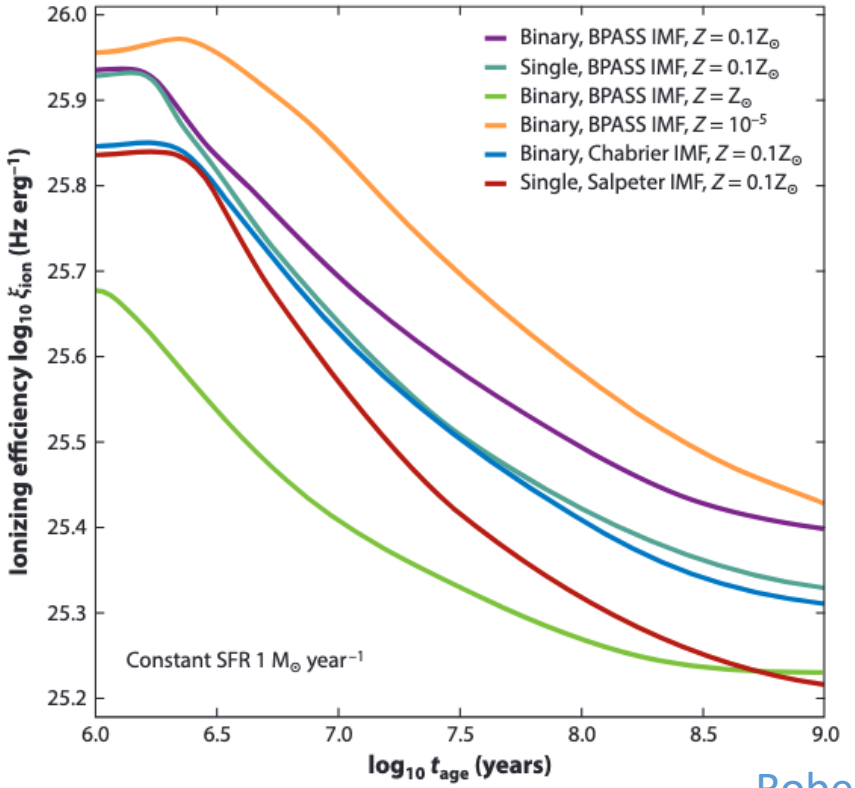
How can we tell if it's early galaxies?

$$\dot{n}_\gamma = f_{esc}\xi_{ion}\rho_{UV}$$

Where ρ_{UV} is the total UV luminosity density, f_{esc} is the fraction of Lyman continuum photons (< 91.2 nm) that escape galaxies, and ξ_{ion} is the ionizing photon production efficient that gives the LyC photons per unit UV luminosity.



Harikane et al. 2022



Robertson 2022

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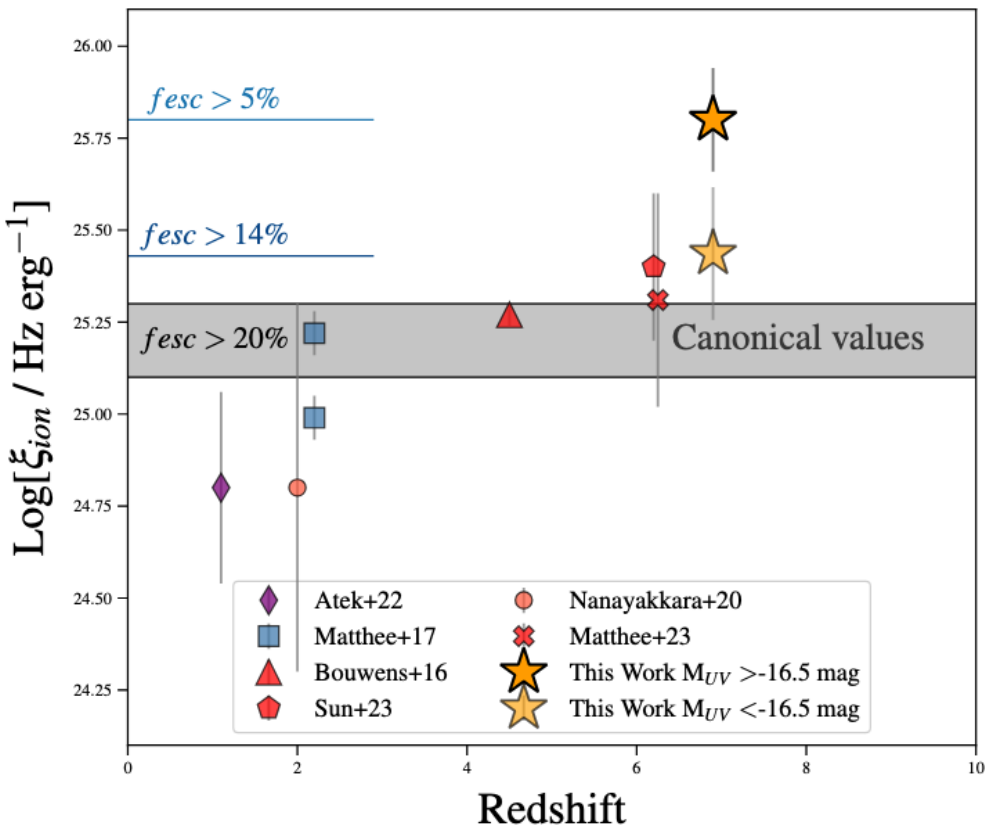
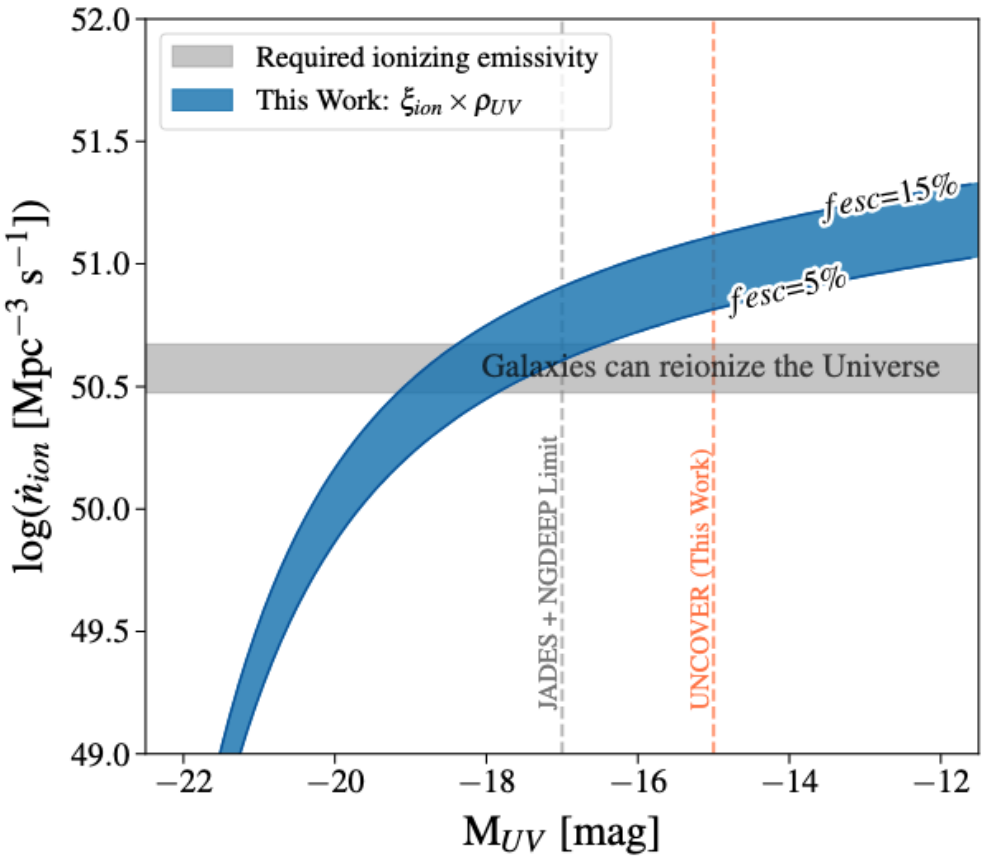


Atek et al 2024



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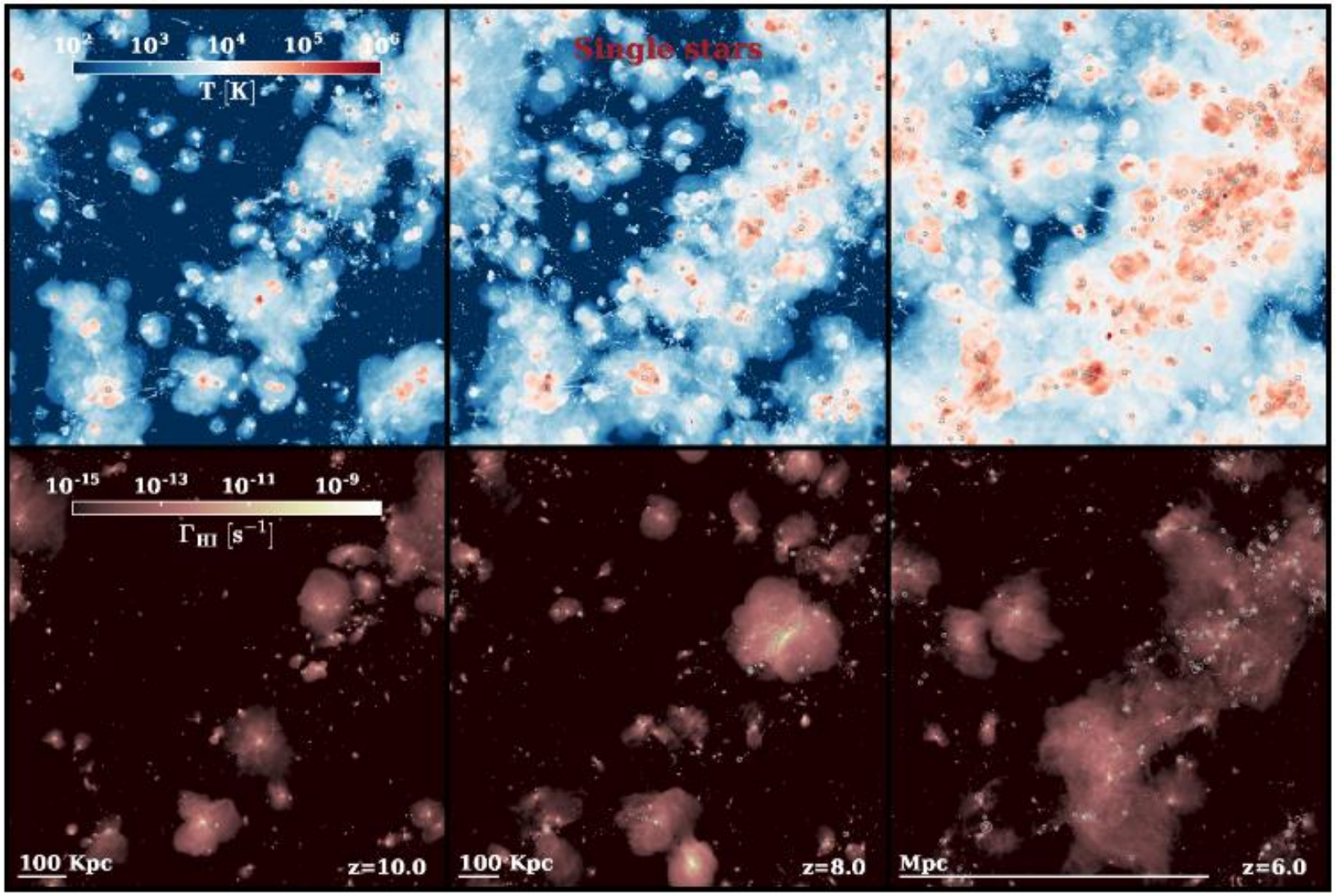
WEBB TELESCOPE FINDS DWARF GALAXIES LIT UP THE EARLY UNIVERSE

BY: COLIN STUART | MARCH 7, 2024 | 0

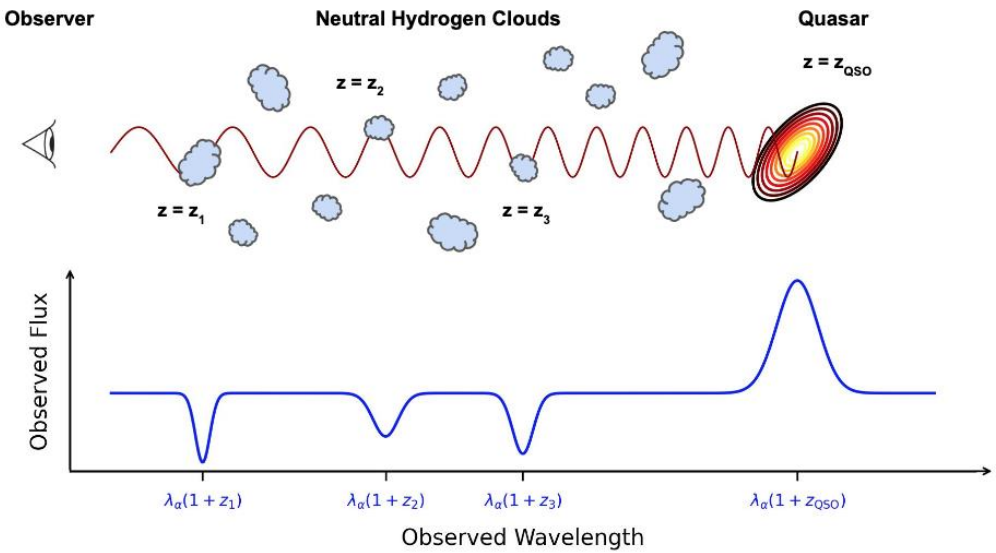
Faint, small galaxies ionized the opaque fog that obscured the early universe.

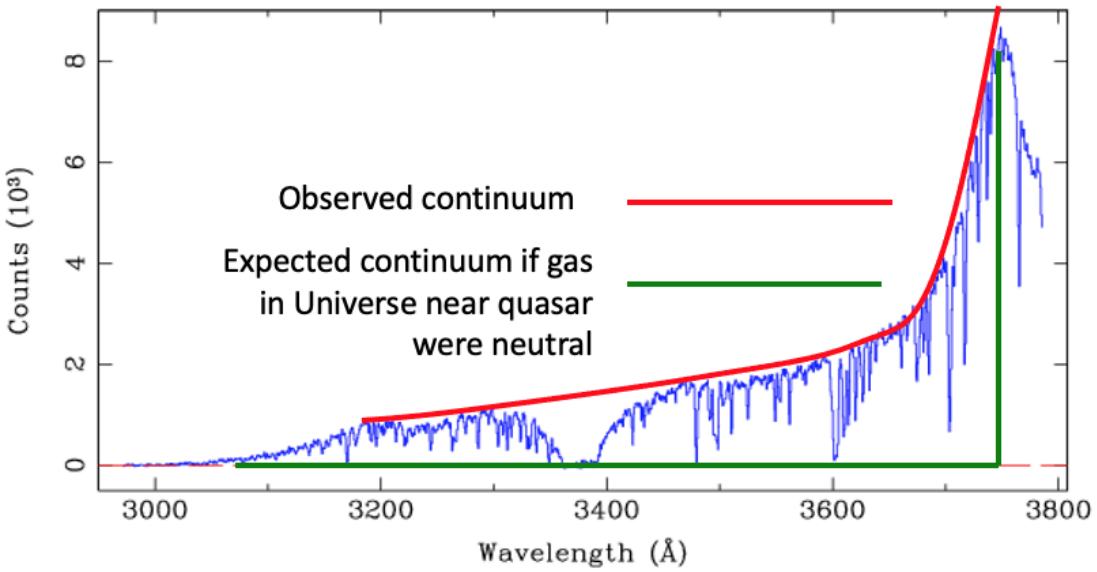
Timing of reionization



Rosdahl et al 2021

Timing of reionization – Gunn Peterson Trough





$$\tau_{GP} = \int n_{HI}(z)\sigma_\alpha ds$$

τ_{GP} is the Gunn-Peterson optical depth, n_{HI} is the neutral hydrogen number density, σ_α is the Lyman-α cross-section, ds is the proper distance element.

Given that $n_{HI} = x_{HI} n_H$, for our universe:

$$\tau_{GP}(z) = 3.8 \times 10^4 x_{HI}(1+z)^{3/2}$$

So if the universe was fully neutral ($x_{HI} = 1$) then

$\tau_{GP}(z = 6) \sim 10^5$ (Complete absorption!)

But even $x_{HI} = 10^{-4}$,

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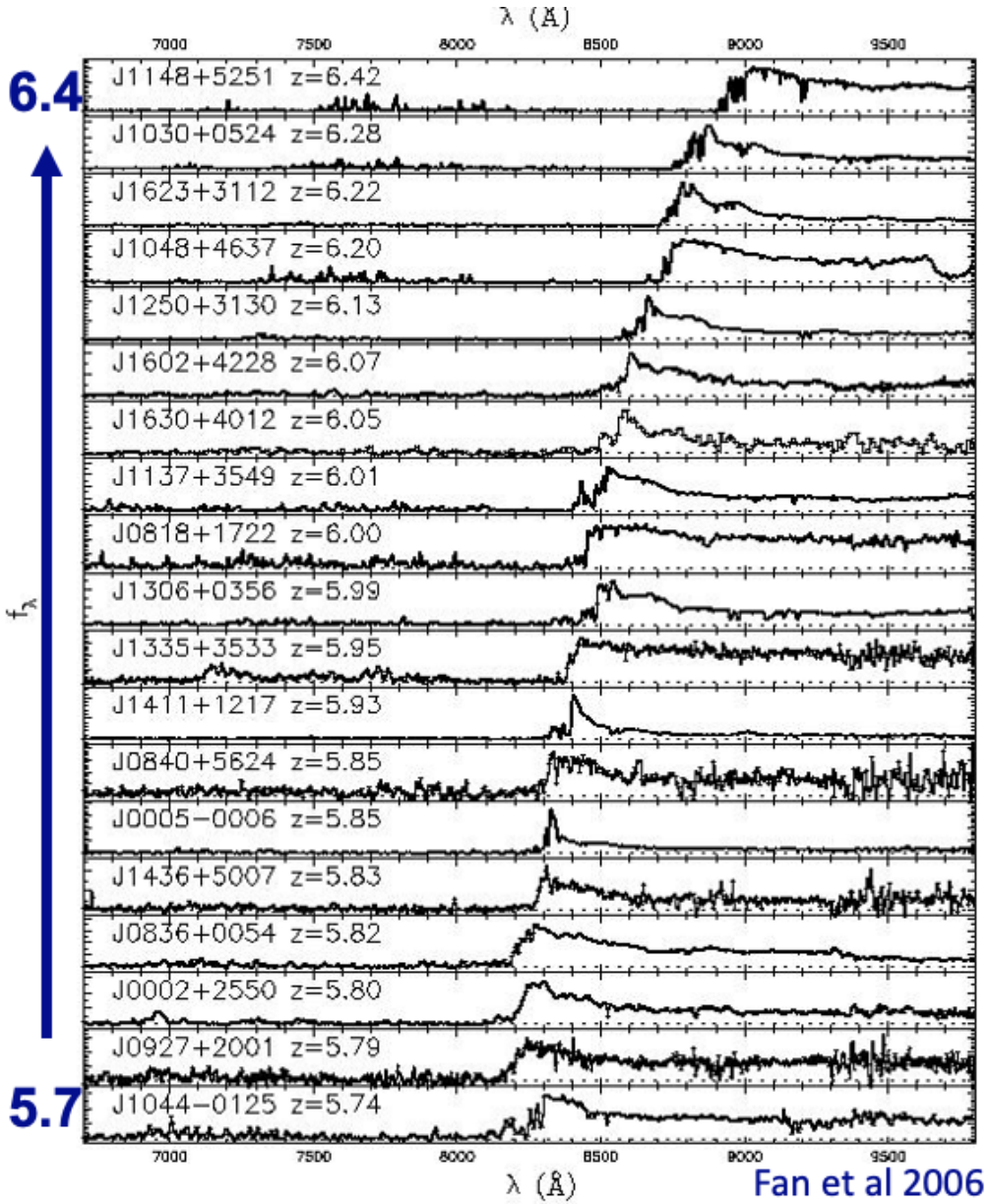
So, even a tiny neutral fraction leads to strong absorption

Timing of reionization – Gunn Peterson Trough



Fan et al 2006

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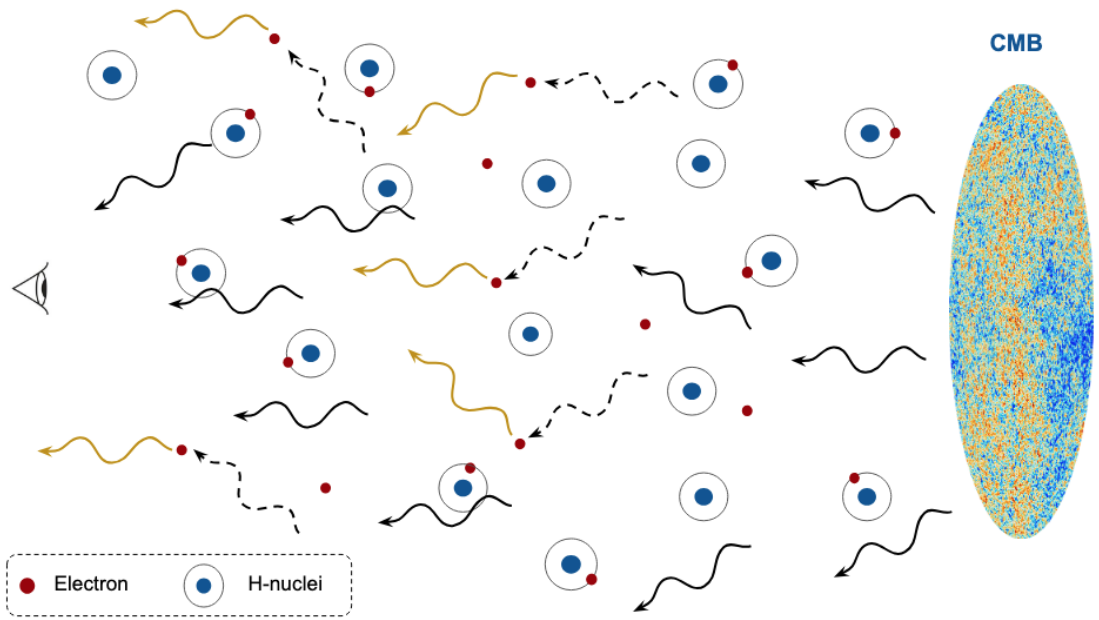
Timing of reionization – CMB

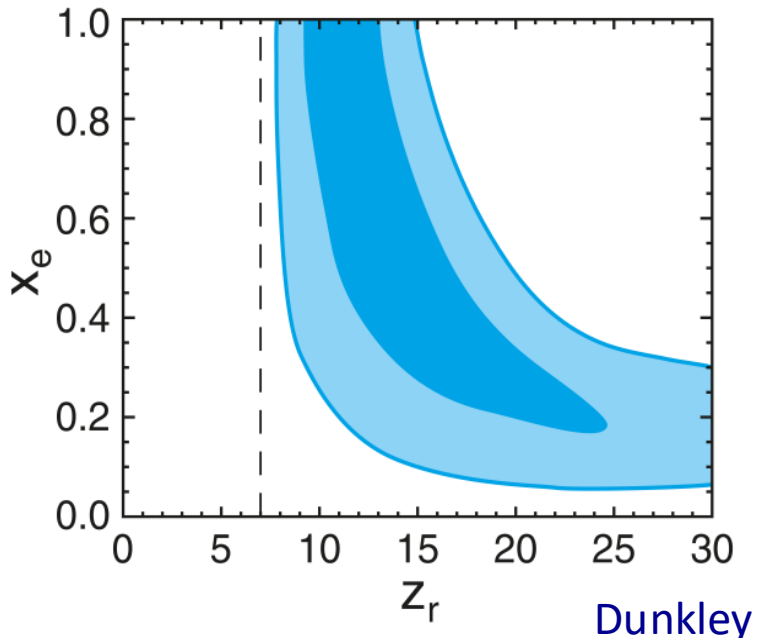


The CMB gives a constraint on reionization because of Thomson scattering of CMB photons on free electrons in the ionized IGM. This Thomson optical depth gives an 'integrated' constraint on the timing.

$$\tau_e = \int_z^{z_{reion}} c\,\sigma_T\, n_e(z) \frac{dz}{(1+z)H(z)}$$

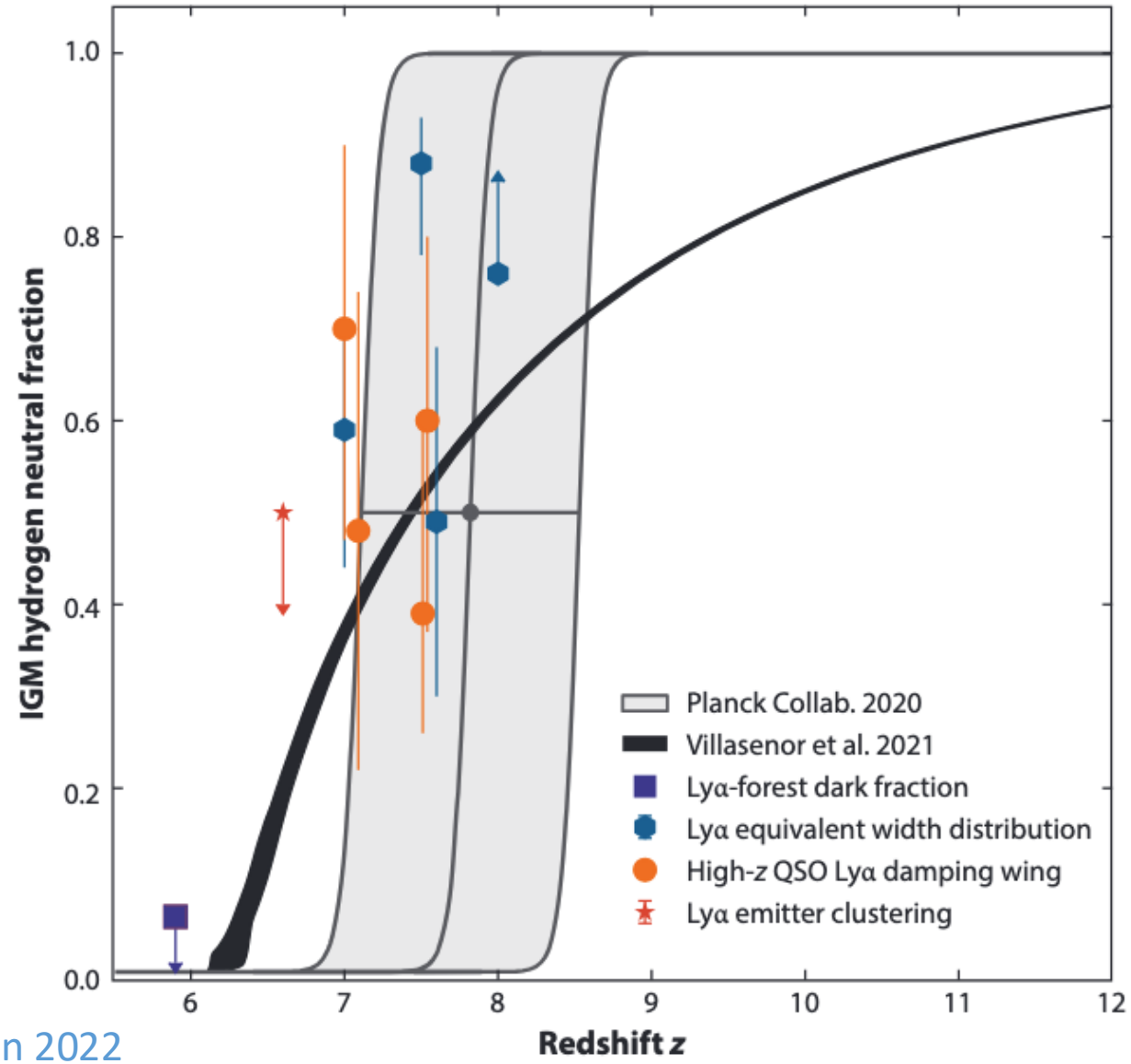
τ_e is Thomson optical depth, σ_T is the Thomson cross-section ($6.652 \times 10^{-29} m^2$), $n_e(z)$ is the electron number density. This is actually measured as a polarization, and is difficult.

- This limit rules out a high ionisation fraction at z > 15.
- But it actually *requires* some ionisation beyond z=6.



Dunkley et al. 2008

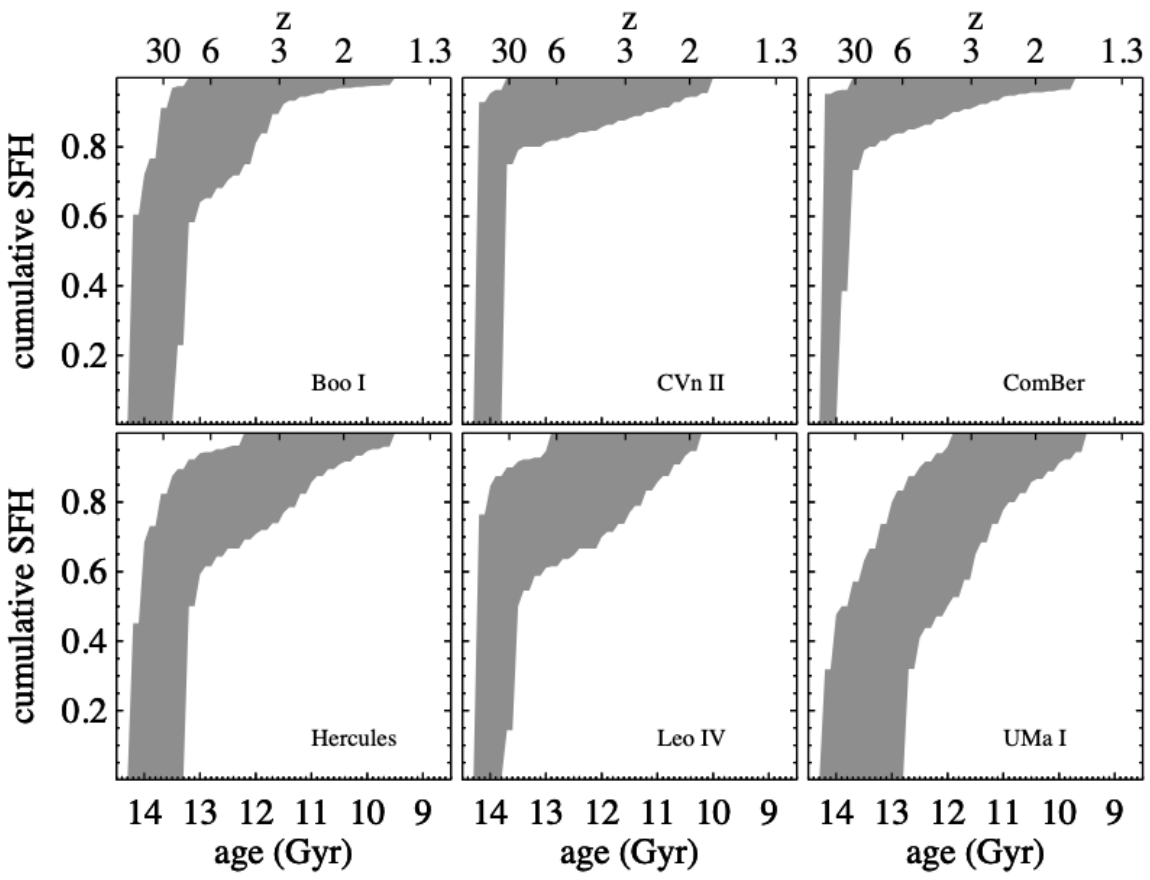
Combined constraints on reionization



Robertson 2022

In the hopefully near future 21-cm emission measures in the Radio will give a whole new window on reionzation

Quenching of small galaxies



Brown et al 2014

Photoheating feedback raises the temperature of the IGM, preventing low-mass halos from accreting gas and forming stars. The temperature of the photoheated gas after reionization is $T \sim 10^4 K$.

A dark matter halo can only retain gas if its virial temperature exceeds the temperature of the reionized IGM. The virial temperature can be

$$T_{vir} = 1.98 \times 10^4 \left(\frac{M_h}{10^8 M_\odot}\right)^{\frac{2}{3}} \left(\frac{1+z}{10}\right) K$$

At z = 6, $M_H \approx 10^9 M_\odot$ while at z = 10, this is lower $M_H \approx 3 \times 10^8 M_\odot$

Thus, galaxies in halos below $\approx 10^9 M_\odot$ are strongly suppressed post-reionization